

# **API Security Risk Analysis Report**

Cyber Security Task 3 (2026)

Future Interns

## **Submitted By**

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Task: API Security Risk Analysis

## Introduction

APIs (Application Programming Interfaces) are used in modern applications to connect services like mobile apps, websites, and dashboards. If APIs are not secured properly, attackers can access sensitive data, bypass authentication, and misuse system resources. This report analyzes a public API to identify common security risks and understand how API security works in real systems.

## API Tested

API Used:

<https://jsonplaceholder.typicode.com>

This is a public test API used for learning and development purposes and does not contain real user-sensitive data.

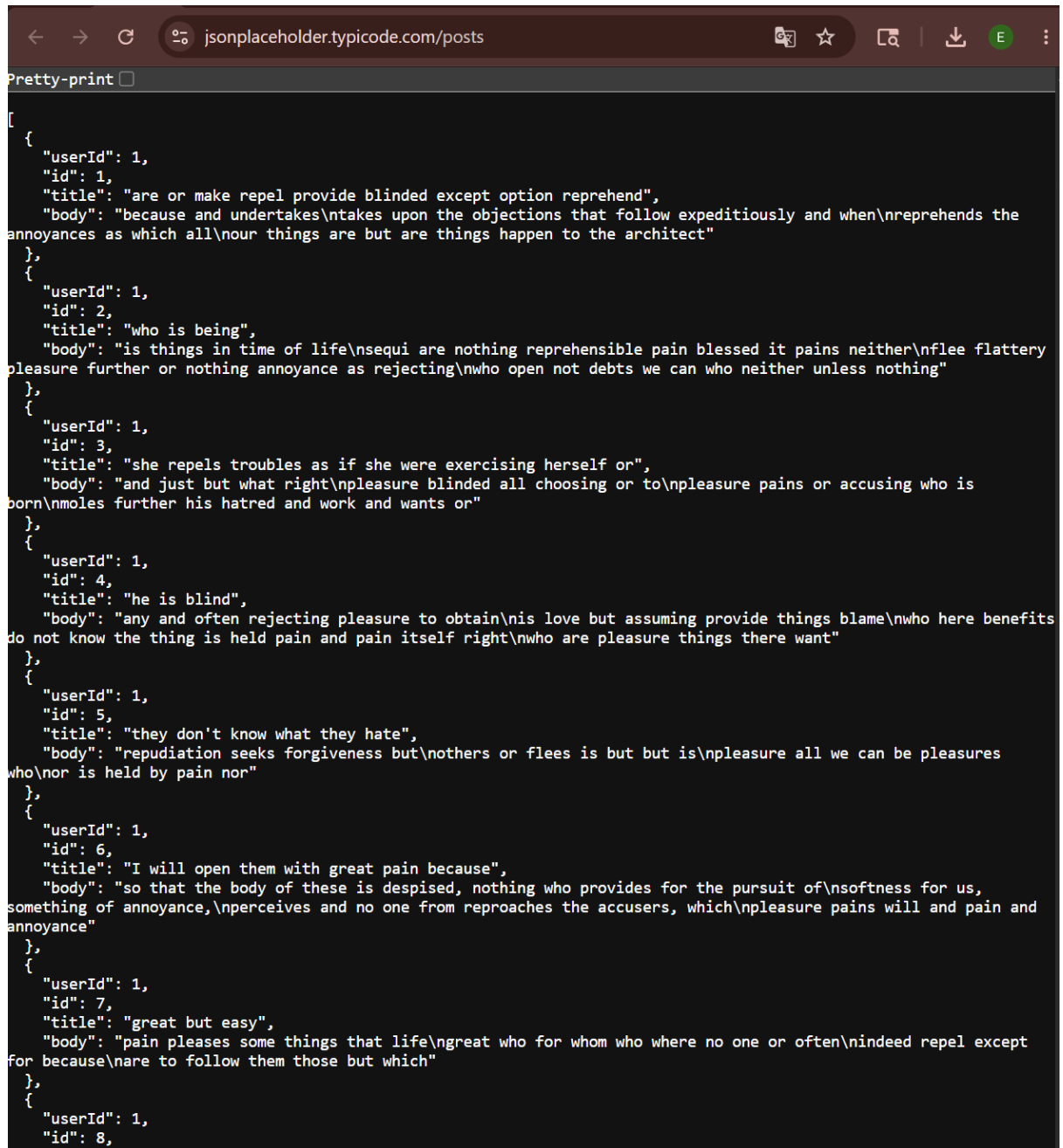
## Endpoints Tested

The following endpoints were analyzed:

- /posts
- /users
- /comments

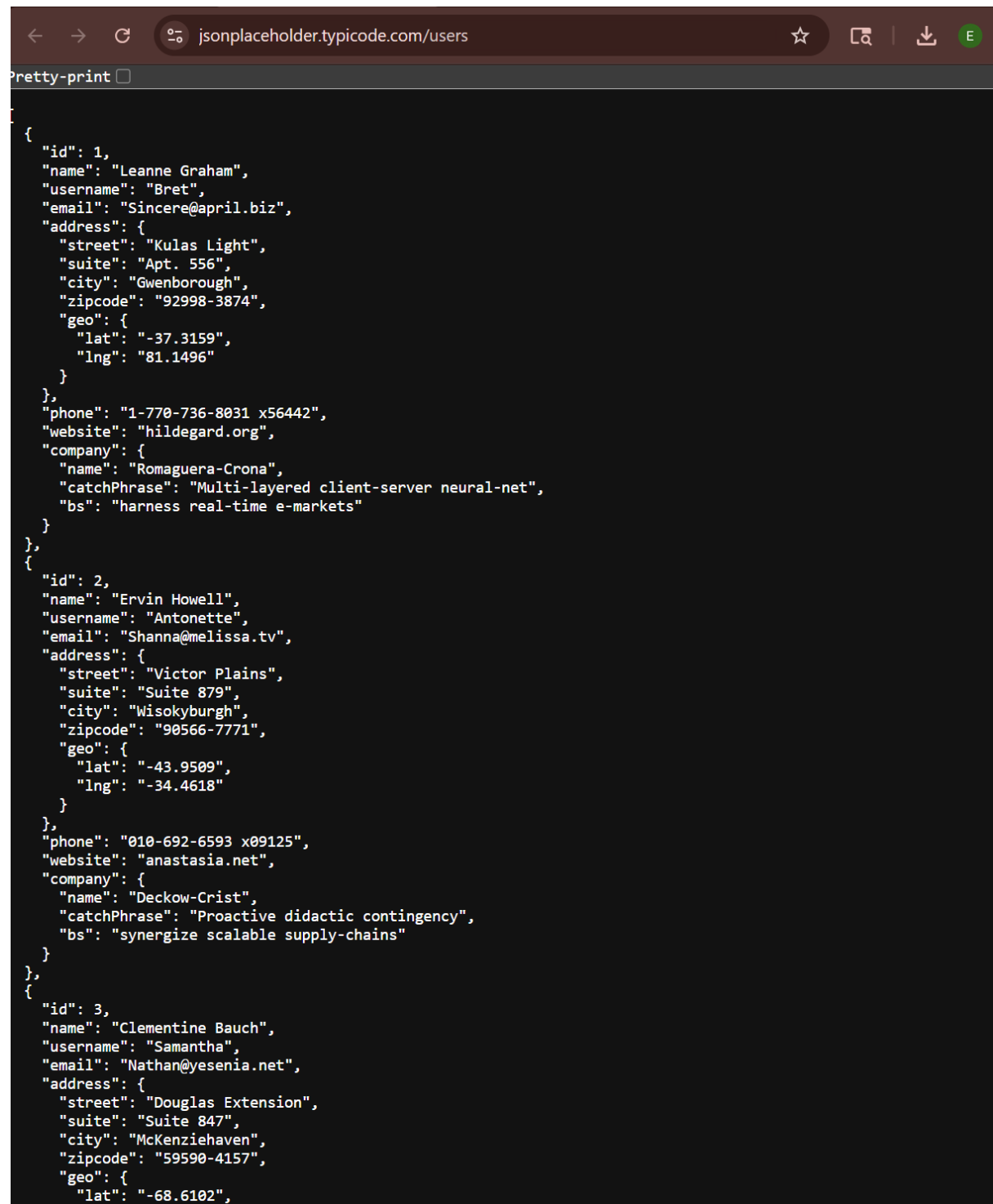
# API Testing Evidence

## 1. API Open Screenshot



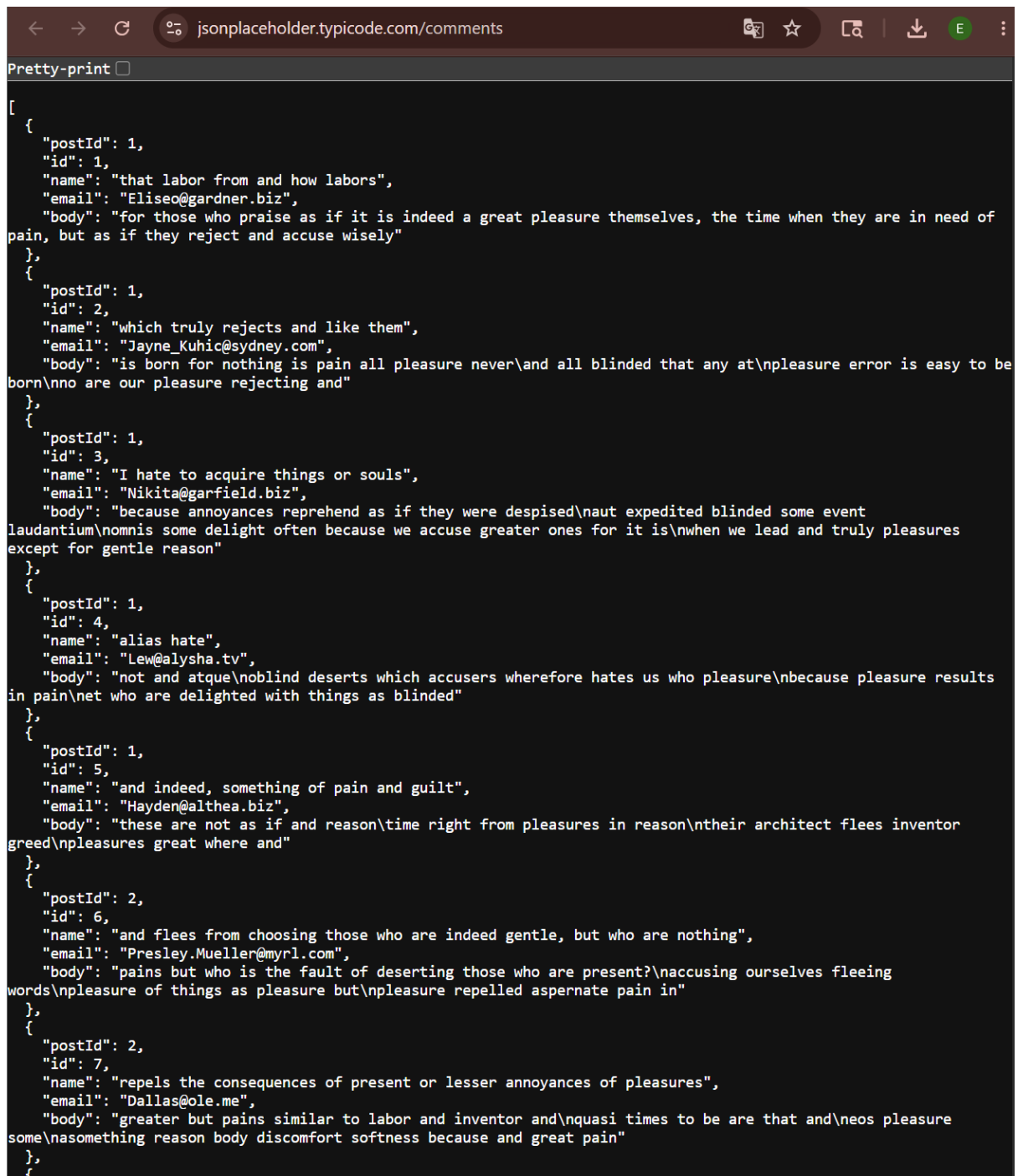
```
[
  {
    "userId": 1,
    "id": 1,
    "title": "are or make repel provide blinded except option reprehend",
    "body": "because and undertakes\ntakes upon the objections that follow expeditiously and when\nreprehends the annoyances as which all\nour things are but are things happen to the architect"
  },
  {
    "userId": 1,
    "id": 2,
    "title": "who is being",
    "body": "is things in time of life\nsequi are nothing reprehensible pain blessed it pains neither\nflee flattery pleasure further or nothing annoyance as rejecting\nwho open not debts we can who neither unless nothing"
  },
  {
    "userId": 1,
    "id": 3,
    "title": "she repels troubles as if she were exercising herself or",
    "body": "and just but what right\npleasure blinded all choosing or to\npleasure pains or accusing who is born\nmoles further his hatred and work and wants or"
  },
  {
    "userId": 1,
    "id": 4,
    "title": "he is blind",
    "body": "any and often rejecting pleasure to obtain\nis love but assuming provide things blame\nwho here benefits do not know the thing is held pain and pain itself right\nwho are pleasure things there want"
  },
  {
    "userId": 1,
    "id": 5,
    "title": "they don't know what they hate",
    "body": "repudiation seeks forgiveness but\nothers or flees is but but is\npleasure all we can be pleasures who\nnor is held by pain nor"
  },
  {
    "userId": 1,
    "id": 6,
    "title": "I will open them with great pain because",
    "body": "so that the body of these is despised, nothing who provides for the pursuit of\nsoftness for us, something of annoyance,\nperceives and no one from reproaches the accusers, which\npleasure pains will and pain and annoyance"
  },
  {
    "userId": 1,
    "id": 7,
    "title": "great but easy",
    "body": "pain pleases some things that life\ngreat who for whom who where no one or often\nindeed repel except for because\nare to follow them those but which"
  },
  {
    "userId": 1,
    "id": 8,
    "title": "great but easy",
    "body": "pain pleases some things that life\ngreat who for whom who where no one or often\nindeed repel except for because\nare to follow them those but which"
  }
]
```

## 2. Users Endpoint Screenshot



```
{
  "id": 1,
  "name": "Leanne Graham",
  "username": "Bret",
  "email": "Sincere@april.biz",
  "address": {
    "street": "Kulas Light",
    "suite": "Apt. 556",
    "city": "Gwenborough",
    "zipcode": "92998-3874",
    "geo": {
      "lat": "-37.3159",
      "lng": "81.1496"
    }
  },
  "phone": "1-770-736-8031 x56442",
  "website": "hildegard.org",
  "company": {
    "name": "Romaguera-Crona",
    "catchPhrase": "Multi-layered client-server neural-net",
    "bs": "harness real-time e-markets"
  }
},
{
  "id": 2,
  "name": "Ervin Howell",
  "username": "Antonette",
  "email": "Shanna@melissa.tv",
  "address": {
    "street": "Victor Plains",
    "suite": "Suite 879",
    "city": "Wisokyburgh",
    "zipcode": "90566-7771",
    "geo": {
      "lat": "-43.9509",
      "lng": "-34.4618"
    }
  },
  "phone": "010-692-6593 x09125",
  "website": "anastasia.net",
  "company": {
    "name": "Deckow-Crist",
    "catchPhrase": "Proactive didactic contingency",
    "bs": "synergize scalable supply-chains"
  }
},
{
  "id": 3,
  "name": "Clementine Bauch",
  "username": "Samantha",
  "email": "Nathan@yesenia.net",
  "address": {
    "street": "Douglas Extension",
    "suite": "Suite 847",
    "city": "McKenziehaven",
    "zipcode": "59590-4157",
    "geo": {
      "lat": "-68.6102",
      "lng": "-87.7163"
    }
  },
  "phone": "1-462-339-3932 x65430",
  "website": "kathleen.biz",
  "company": {
    "name": "Frisch-Costa",
    "catchPhrase": "Proactive dynamic neural-net",
    "bs": "synthesize real-time e-markets"
  }
}
```

## 3. Comments Endpoint



```
[
  {
    "postId": 1,
    "id": 1,
    "name": "that labor from and how labors",
    "email": "Eliseo@gardner.biz",
    "body": "for those who praise as if it is indeed a great pleasure themselves, the time when they are in need of pain, but as if they reject and accuse wisely"
  },
  {
    "postId": 1,
    "id": 2,
    "name": "which truly rejects and like them",
    "email": "Jayne_Kuhic@sydney.com",
    "body": "is born for nothing is pain all pleasure never\\and all blinded that any at\\npleasure error is easy to be born\\nno are our pleasure rejecting and"
  },
  {
    "postId": 1,
    "id": 3,
    "name": "I hate to acquire things or souls",
    "email": "Nikita@garfield.biz",
    "body": "because annoyances reprehend as if they were despised\\naut expedited blinded some event laudantium\\nomnis some delight often because we accuse greater ones for it is\\nwhen we lead and truly pleasures except for gentle reason"
  },
  {
    "postId": 1,
    "id": 4,
    "name": "alias hate",
    "email": "Lew@alysha.tv",
    "body": "not and atque\\noblind deserts which accusers wherefore hates us who pleasure\\nbecause pleasure results in pain\\net who are delighted with things as blinded"
  },
  {
    "postId": 1,
    "id": 5,
    "name": "and indeed, something of pain and guilt",
    "email": "Hayden@althea.biz",
    "body": "these are not as if and reason\\time right from pleasures in reason\\ntheir architect flees inventor greed\\npleasures great where and"
  },
  {
    "postId": 2,
    "id": 6,
    "name": "and flees from choosing those who are indeed gentle, but who are nothing",
    "email": "Presley.Mueller@myrl.com",
    "body": "pains but who is the fault of deserting those who are present?\\naccusing ourselves fleeing words\\npleasure of things as pleasure but\\npleasure repelled aspernate pain in"
  },
  {
    "postId": 2,
    "id": 7,
    "name": "repels the consequences of present or lesser annoyances of pleasures",
    "email": "Dallas@ole.me",
    "body": "greater but pains similar to labor and inventor and\\nquasi times to be are that and\\neos pleasure some\\nasomething reason body discomfort softness because and great pain"
  },
  {
    "postId": 2,
    "id": 8,
    "name": "repels the consequences of present or lesser annoyances of pleasures",
    "email": "Dallas@ole.me",
    "body": "greater but pains similar to labor and inventor and\\nquasi times to be are that and\\neos pleasure some\\nasomething reason body discomfort softness because and great pain"
  },
  {
    "postId": 2,
    "id": 9,
    "name": "repels the consequences of present or lesser annoyances of pleasures",
    "email": "Dallas@ole.me",
    "body": "greater but pains similar to labor and inventor and\\nquasi times to be are that and\\neos pleasure some\\nasomething reason body discomfort softness because and great pain"
  },
  {
    "postId": 2,
    "id": 10,
    "name": "repels the consequences of present or lesser annoyances of pleasures",
    "email": "Dallas@ole.me",
    "body": "greater but pains similar to labor and inventor and\\nquasi times to be are that and\\neos pleasure some\\nasomething reason body discomfort softness because and great pain"
  }
]
```

## Security Findings

### 1. No Authentication Required

The API can be accessed without login or API keys.

### 2. Public Data Exposure

All data is openly accessible to any user.

### **3. No Access Control**

No restrictions on who can view data.

### **4. No Rate Limiting**

Multiple requests can be made without restriction.

## **Risk Classification**

| <b>Risk</b>       | <b>Severity</b> |
|-------------------|-----------------|
| No Authentication | High            |
| Data Exposure     | High            |
| No Access Control | High            |
| No Rate Limiting  | Medium          |

## **Business Impact**

If such an API is used in a real application, attackers could easily access sensitive user information, leading to data leaks, privacy issues, and system misuse. This can damage user trust and organization reputation.

## **Recommendations**

- Implement authentication (JWT / OAuth)
- Add authorization controls (RBAC)

- Enable rate limiting to prevent abuse
- Validate all input data
- Encrypt sensitive information
- Monitor API usage logs

## Conclusion

API security is a critical part of modern application development. This analysis shows that lack of authentication and access control can lead to serious data exposure risks. Proper security mechanisms must be implemented to protect APIs from unauthorized access and misuse.

End of Report